

1. Modify syscall.h

add this lines:

```
#define SYS_numvp 30
#define SYS_numpp 31
#define SYS_getptsize 32
```

2. Declare in user.h

add this lines:

```
int numvp(void);
int numpp(void);
int getptsize(void);
```

3. Implement in sysproc.c

add this code:

```
extern pte_t* walkpgdir(pde_t *pgdir, const void *va, int alloc);
```

```
int sys_numvp(void) {
    struct proc *p = myproc();
    return (p->sz + PGSIZE - 1)/PGSIZE + 1;
}
```

```
int sys_numpp(void) {
    struct proc *p = myproc();
    pte_t *pte;
    int count = 0;
    for(uint a = 0; a < p->sz; a += PGSIZE){
        pte = walkpgdir(p->pgdir, (void*)a, 0);
        if(pte && (*pte & PTE_P)) count++;
    }
    return count;
}
```

```
int sys_getptsize(void) {
    struct proc *p = myproc();
    int count = 1; // outer page directory
    for(int i=0; i<NPENTRIES; i++){
        if(p->pgdir[i] & PTE_P) count++;
    }
    return count;
}
```

4. Modify syscall.c

add this lines:

```
extern int sys_numvp(void);
extern int sys_numpp(void);
```

```
extern int sys_getpagesize(void);
```

```
[SYS_numvp] sys_numvp,  
[SYS_numpp] sys_numpp,  
[SYS_getpagesize] sys_getpagesize,
```

5. edit usys.S

```
SYSCALL(numvp)  
SYSCALL(numpp)  
SYSCALL(getpagesize)
```

6. Create memtest.c

add this code:

```
#include "types.h"  
#include "stat.h"  
#include "user.h"  
int main(void) {  
    printf(1, "Virtual pages: %d \n", numvp());  
    printf(1, "Physical pages: %d \n", numpp());  
    printf(1, "Page table pages: %d \n", getpagesize());  
    exit();  
}
```

7. Edit Makefile

add this code:

```
_memtest
```

Machine View

SeaBIOS (version 1.16.3-debian-1.16.3-2)

iPXE (<https://ipxe.org>) 00:03.0 CA00 PCI2.10 PnP PMM+1EFCB050+1EF0B050 CA00

Booting from Hard Disk...

cpu0: starting 0

sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58

init: starting sh

12340920\$memtest

Virtual pages: 4

Physical pages: 3

Page table pages: 66

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